Fin-Cash Test Plan

Capstone 2025–2026 | Team 13 | Updated to include the additional financial features

1. Introduction

This plan defines how the Fin-Cash team will verify that the system meets the Software Requirements Specification (SRS). Fin-Cash is a web-based platform that imports financial data from CSV and XLSX files, generates reports, displays KPI dashboards, forecasts trends and controls access by role (RBAC). This updated plan also covers testing for Smart Budget Monitoring, Quick Search & Filter, Monthly Financial Summary and AI-powered financial features.

2. Scope

In scope:

- Secure login and Role-Based Access Control (RBAC), including admin management of accounts, roles and permissions.
- Importing and validating financial data from CSV and XLSX files.
- Processing and aggregating data using predefined calculation rules.
- Interactive dashboards showing key financial KPIs.
- Automated reports: income statements, balance sheets and cash flow reports.
- Financial forecasting from historical data.
- Exporting reports and analytics to PDF, XLSX and CSV.
- Activity logging (audit log) for security.
- Non-functional checks: 3-second dashboard load, TLS/AES-256 encryption, 99.5% uptime, daily backup and browser compatibility (Chrome, Edge and Firefox).
- Smart Budget Monitoring: verify budget usage percentages, warning thresholds and projected budget overruns.
- Quick Search & Filter: verify transaction search by name and filtering by Revenue or Expense.
- Monthly Financial Summary: verify monthly total revenue, total expenses and net profit, including month selection.
- AI Spending Anomaly Detection: verify that unusual expenses are flagged for review based on historical spending patterns.
- AI Financial Summary: verify that generated summaries accurately describe key changes in the financial data.
- AI Expense Forecasting: verify that forecasts use historical data and display understandable estimates.

Out of scope:

- Direct banking transactions.
- Replacing existing accounting systems.
- Integration with external legacy databases (a possible future enhancement).
- Testing the corporate IT infrastructure, network and third-party services.

3. Test Strategy

Testing follows the activities defined in the SRS. Each test uses sample data supplied by the project team. Defects are logged, fixed and retested.

- **Functional testing:** verify data import and validation, report generation, forecasting, export and audit logging against written test cases.
- **Security and access control testing:** confirm each role (finance staff, analyst, manager/executive and administrator) sees only permitted features and data, and that unauthorized users cannot export restricted reports.
- **Usability testing:** check dashboard clarity, ease of navigation, search and filter usability, and month selection.
- **Performance and stress testing:** measure dashboard and chart load times with multiple simultaneous users against the 3-second target.

- **Data accuracy verification:** compare summaries and exported files with source datasets; pass criterion is zero calculation discrepancy.
- **Compatibility testing:** run the main flows in Chrome, Edge and Firefox.

3.1 Additional Feature Test Cases

Feature	Test	Expected result
Smart Budget Monitoring	Load a department budget and transactions that use 90% of the budget.	Correct percentage is shown and a warning appears when the configured threshold is reached; projected overrun is clearly identified.
Quick Search & Filter	Search by transaction name, then select Revenue and Expense filters.	Matching transactions appear; each type filter displays only the selected transaction type.
Monthly Financial Summary	Select a month containing known revenue and expense records.	Total revenue, total expenses and net profit match the source records for the selected month.
AI Spending Anomaly Detection	Provide normal historical expenses and one unusually high expense.	The unusual transaction is flagged for review, and normal transactions are not incorrectly reported without reason.
AI Financial Summary	Generate a summary for a dataset with known month-to-month changes.	Summary describes the main changes accurately and does not invent financial values.
AI Expense Forecasting	Run a forecast using historical expense data.	A future estimate is displayed and is based on available historical data; insufficient data is clearly indicated.

4. Environmental Needs

Software requirements:

- Modern web browsers: Google Chrome, Microsoft Edge and Mozilla Firefox.
- Microsoft Excel or similar to prepare and check XLSX and CSV test files.
- Fin-Cash application build and test database loaded with sample financial data, including budgets, dated transactions and historical expense records.
- Test case file and defect tracking sheet.
- Load-testing tool for multi-user performance tests.

Hardware requirements:

- Computers or laptops for testers, able to run a modern browser.
- Server or hosting environment for the Fin-Cash web platform.
- Stable internet or internal network connection.
- Several devices to simulate concurrent users.

5. Staff & Responsibilities

Role	Staff member	Responsibilities
Test Lead	Mohamed Waleed	Owns the test plan, assigns tasks, tracks progress and writes the final test report.
Functional Tester	Marwan Hazem	Tests data import, reports, forecasting, export, search/filter, budget monitoring and monthly summary.
Security Tester	Youseef Mohamed	Tests login, RBAC, admin permissions and audit logging.
Usability and Performance Tester	Sama Adel	Runs usability sessions and dashboard load tests, and checks browser compatibility.
Data Accuracy Tester	Youmna Reda	Compares reports and exports with source data; checks AI summaries and forecast outputs for accuracy and logs defects.

Role	Staff member	Responsibilities
Supervisors	Eng. Mostafa, Eng. Marwa	Guide the team and review the plan and results.

6. Deliverables & Scheduling

Deliverable / Milestone	Start date	Duration	End date	Comments
Test plan and test cases file	11 Oct 2026	5 days	15 Oct 2026	Includes sample data setup and additional feature cases.
Functional and security/RBAC testing	18 Oct 2026	7 days	24 Oct 2026	Includes search/filter, budget checks and monthly summary; defects logged and retested.
Usability, performance and data accuracy testing	25 Oct 2026	7 days	31 Oct 2026	Includes browser compatibility and checks of AI summaries and forecasts.
Final test report and review	1 Nov 2026	3 days	3 Nov 2026	Summarizes test outcomes and unresolved defects.

7. Review and Approvals

Role	Name	Date	Signature
Test Lead	Mohamed Waleed	10 Oct 2026	Mohamed Waleed
Supervisor	Eng. Mostafa		
Supervisor	Eng. Marwa		